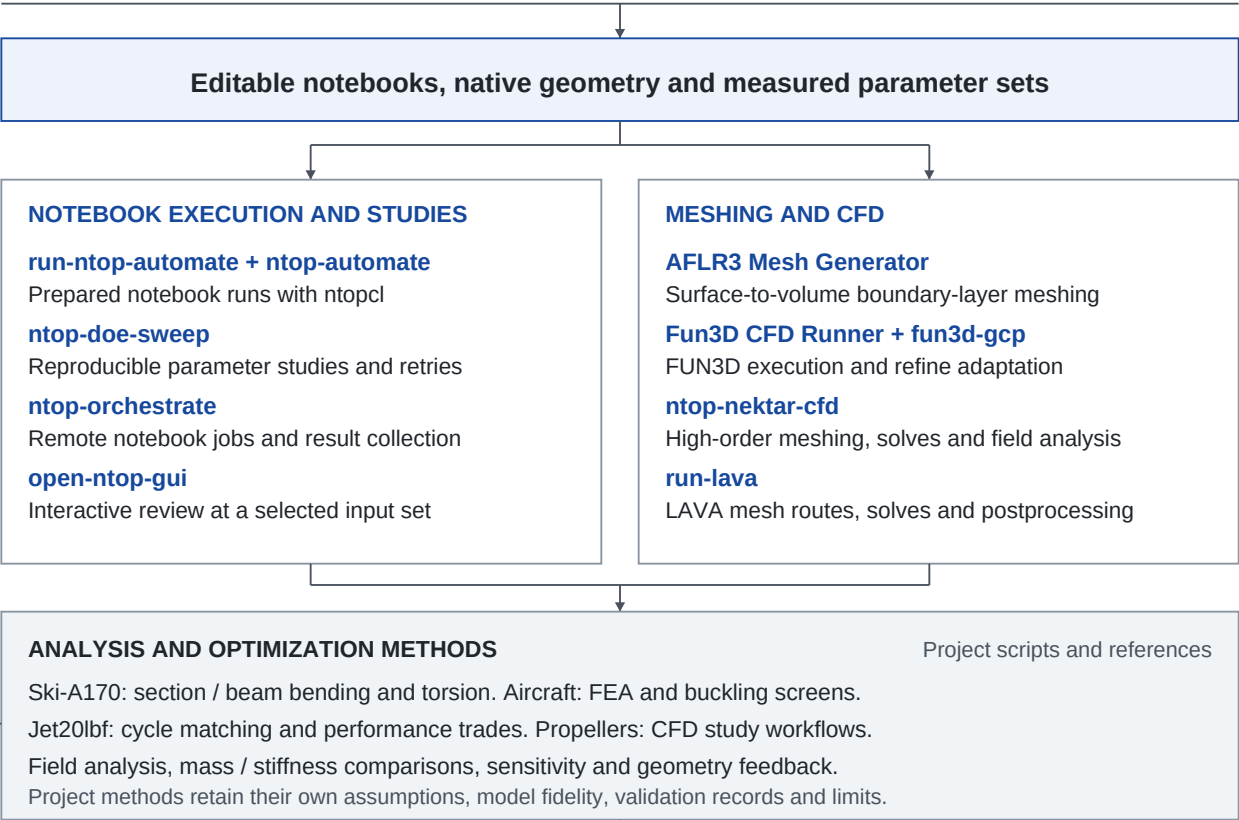
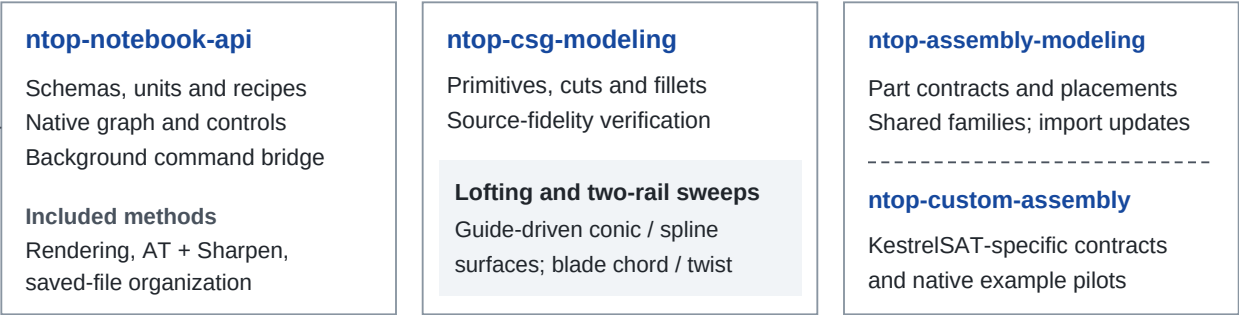


# Current skills for nTop engineering experiments

18 named skills, with lofting and project analysis methods shown explicitly.

## AUTHORING AND MODELING



## POSTPROCESSING AND REPORTING



**Verification at every stage: source, graph, geometry, mesh, solver and result checks.**

Blue names identify skills. Gray method panels identify supporting guidance and project implementations.

# Potential future engineering skills

Ten proposed skill briefs. These are not installed or validated capabilities.

## Shared aircraft design contract

Mission, geometry, interfaces, materials, loads, mass, units and coordinate frames  
Every result retains assumptions, sources, revision identity and uncertainty.

### George Irving configurator

Capture George-reviewed configuration rules.  
Generate coordinated aircraft layouts.  
Expose editable design choices and constraints.

### Expert airframing

Define spars, frames, ribs and longerons.  
Coordinate joints and structural load paths.  
Check access and removable skin panels.

### Propulsion and installation

Match engine or propulsor to the mission.  
Model cycle, power and installed losses.  
Coordinate inlets, exhausts and cooling.

### Weights and mass properties

Maintain component mass and growth ledgers.  
Calculate CG, inertia and loading cases.  
Track uncertainty and configuration changes.

### Stability and control (S&C)

Estimate derivatives, trim and static margins.  
Assess modes and control authority.  
Map limits across the flight / CG envelope.

### Loads and aeroelasticity

Build maneuver, gust and ground load cases.  
Transfer distributed loads into the structure.  
Screen stiffness coupling and flutter risk.

### Structural sizing and joints

Size members, fasteners and connections.  
Check stress, buckling, fatigue and damage.  
Retain allowables and failure-mode coverage.

### Aerodynamics and mission

Build polars and performance models.  
Compare range, endurance and field performance.  
Assess model fidelity and sensitivity.

### Manufacturing and assembly

Evaluate forming, machining and printing routes.  
Define tolerances, tooling and assembly access.  
Estimate process-driven mass, cost and risk.

### Verification and trade studies

Enforce shared constraints across disciplines.  
Run benchmarks and uncertainty studies.  
Rank candidates with traceable evidence.

**Reviewed results update the shared design and can drive new notebook variants.**

Each future skill needs bounded pilots, independent checks and documented operating limits.

Dashed boxes and links indicate proposals. Detailed inputs, outputs and validation gates accompany the figure.